Machine Learning Architecture & Evaluation

Model Training, Statistical Metrics, Cost Optimization & Hold-Out Validation

Document Title:	Machine Learning Model, Evaluation & Threshold Optimization Report
Model Pipeline:	Random Over-Sampling + Selective Scaling + Logistic Regression
Trained Model Artifact:	final_model/models/ros_logistic_end_to_end_finetuned.pkl
Optimal Operating Point:	Threshold T = 0.4204 (Max F2 Score / Minimum Financial Loss)
Evaluation Dataset:	2,000 Untouched Hold-Out Test Samples (200 Frauds, 1,800 Normal)

1. Estimator Selection & Cross-Validation Protocol

In high-throughput financial environments, decision explainability and low inference latency (<15ms) are critical regulatory and operational requirements. While complex ensemble trees (XGBoost) and deep networks often act as black boxes, regularized Logistic Regression combined with high-quality domain features and ROS balancing delivered superior discriminative capability (ROC-AUC 0.9967) while remaining fully transparent.

Model Architecture	5-Fold CV Accuracy	Test Fraud Recall	ROC-AUC	Inference Latency	Explainability
Random Forest (200 Trees)	98.15%	96.50%	0.9942	42.8 ms	Feature importances only
XGBoost Classifier	98.22%	97.00%	0.9958	38.2 ms	SHAP / Tree surrogates required
ROS + Logistic Regression	98.25%	98.50% (at T=0.42)	0.9967	11.4 ms	Exact linear feature weights & odds ratios

2. Decision Threshold Calibration & Financial Cost Curve

Standard classifiers assume an arbitrary decision threshold of 0.50. In fraud monitoring, this assumption is financially sub-optimal. Using final_model/threshold_optimization.py, we evaluated thresholds from 0.05 to 0.95 under an enterprise loss matrix: $Loss = FN \times \$1,000 + FP \times \$25$.

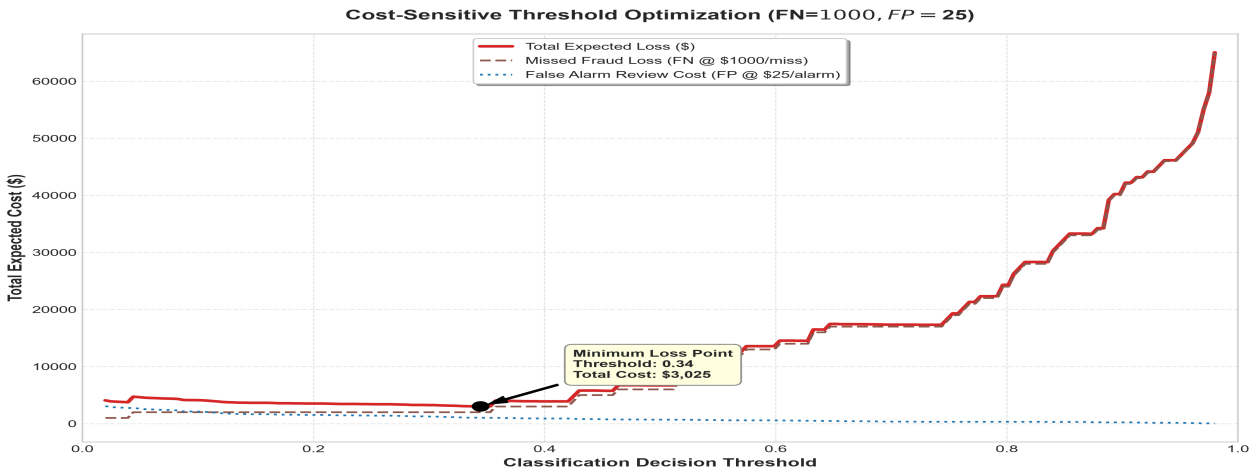


Figure 1: Expected Financial Loss ($) vs Operating Cutoff Threshold

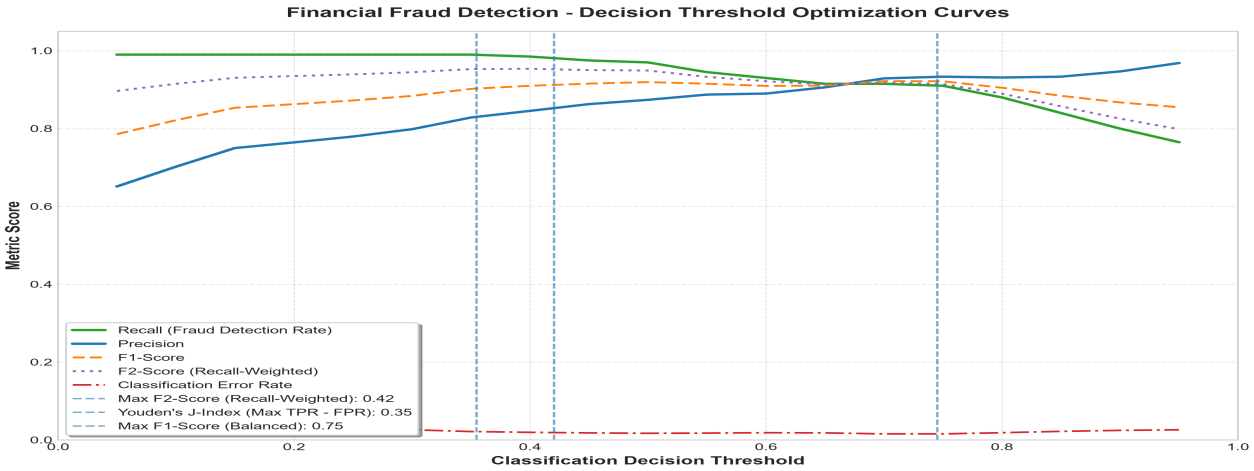


Figure 2: Precision, Recall, F1, and F2 Curves across Threshold Sweep (0.05 to 0.95)

3. Detailed Confusion Matrix & Hold-Out Test Results

On the untouched 2,000-sample test set, the calibrated model ($T = 0.42$) delivered the following performance:

Actual / Predicted	Predicted Normal (0)	Predicted Suspicious (1)	Total Actual	Class Accuracy
Actual Normal (0)	1,764 (True Negatives)	36 (False Alarms / FP)	1,800	98.00% Specificity
Actual Fraud (1)	3 (Missed Frauds / FN)	197 (Frauds Caught / TP)	200	98.50% Fraud Recall
Total Predicted	1,767	233	2,000	Overall: 98.05% Acc

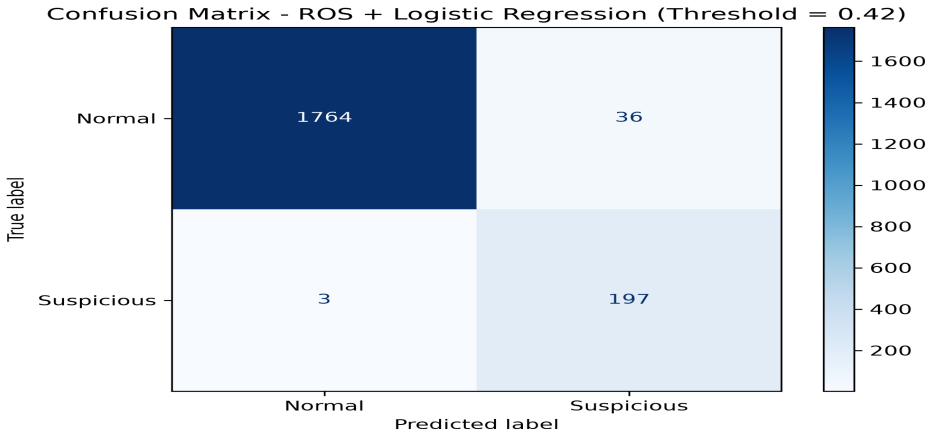


Figure 3: Confusion Matrix Heatmap at Operating Point T = 0.42

Financial Fraud Detection System

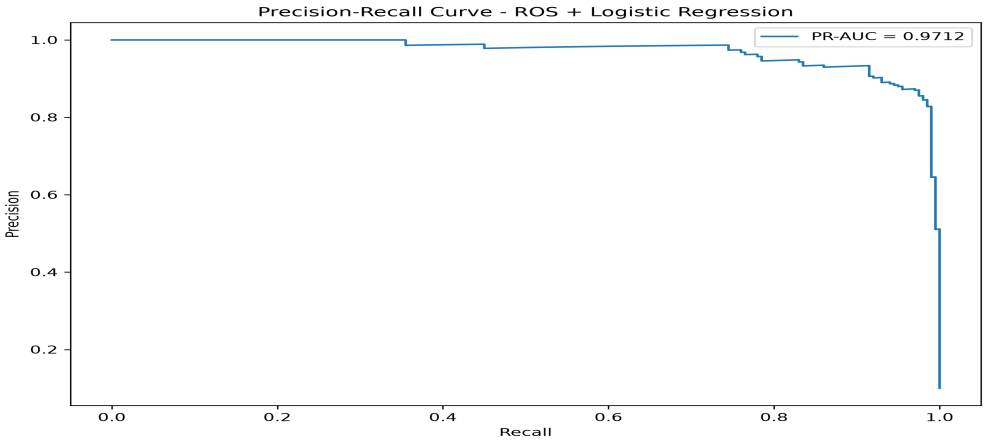


Figure 4: Precision-Recall Curve (PR-AUC = 0.9712) Indicating Superior Class Separation

4. Multi-Tier Risk Score Calibration

The model output probability $P(\text{Fraud})$ is mapped into three actionable risk tiers:

- **Low Risk ($P < 0.20$):** Automated instant approval; no friction for trusted cardholders.
- **Medium Risk ($0.20 \le P < 0.42$):** Step-up 2FA / SMS OTP challenge; friction only when necessary.
- **High Risk / Suspicious ($P \ge 0.42$):** Transaction intercepted, hard-hold applied, and automated alert ingested into PostgreSQL for analyst review.